Numerical Simulation of the Flapping Eagle, Owl and S1223 Airfoils

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***Abstract*—In this study, the aerodynamic performance of the flapping owl, eagle and S1223 airfoils was investigated numerically using ANSYS Fluent commercial CFD code. The study was carried out by solving Reynolds Averaged Navier Stoke (RANS) equations where the turbulence was modeled by k-Epsilon turbulence model. The moving mesh strategy was adapted for the airfoil flapping. The numerical results were obtained for a Reynolds number of 7×10^5 and a flapping rate of $\mp 0.5 \frac{rad}{s}$. The results were presented in terms of pressure contours, streamlines, time history fluctuation of the lift and drag coefficients and aerodynamic efficiency. The downstroke and upstroke pressure contours and streamlines show similar aerodynamic behavior for the studied airfoils. The only differences are the size and the location of vortices. In comparison to the S1223 airfoil, the results showed that the eagle and owl airfoils generated more lift during the downstroke mode and less lift during upstroke mode. In terms of aerodynamic efficiency, eagle's airfoil generates the highest lift to drag ratio followed by S1223 airfoil. The obtained results showed good agreement with the published results in terms of the time history fluctuation trend for the lift and drag coefficients.**

Keywords— aerodynamics, Owl airfoil, eagle airfoil, S1223, flapping, CFD

I. Introduction

Birds use different flapping modes in hovering and forward flight to generate lift and thrust. The bird's airfoils during flapping mode create forward and upward force and thrust in addition to the lift. The flapping in the birds is done in upstroke and downstroke modes which generate most of the aerodynamic forces. In comparison to the fixed airfoil aerodynamics, the flapping airfoils face two major complications, mainly the unsteadiness of the flow and structural movement. Lessons can be learned by studying the flapping birds' airfoil. Those lessons can used by airfoil designers to improve the aerodynamic performance of the low Reynolds number man-made bird's like airfoils such as airfoils used for unmanned aerial vehicle (UAV) and micro aerial vehicle (MAV) wings [1, 2]. Recently, the research in the area of the flapping bio-airfoils aerodynamics gained a lot of popularity, especially in the field of the capability of the numerical simulation in capturing accurately the flow features and aerodynamics forces around flapping airfoils. Shyy et al. [3] Observed biological flight vehicles. They highlighted that the wing motion and flexible airfoils are two key attributes for flight at low Reynolds number flows. They found that the small size of MAVs corresponds in nature to small birds, which do not glide like large birds, but instead flap with considerable change of wing shape during a single flapping cycle. They observed that with flapping and flexible wings, birds overcome the deteriorating aerodynamic performance under steady flow conditions by employing unsteady mechanisms. The numerical study of the flapping owl like airfoil was carried out by Lang et al. [4] to investigate its flapping kinematics using Fluent CFD code. They found that the owl-like airfoil can enhance lift during the downstroke flapping mode. Ashraf et al. [5] numerical simulation investigated the aerodynamic performance of number of unflapping birds' airfoils and compared them to bird like airfoil S1223. They concluded that the eagle and stork airfoils performed aerodynamically better than S1223 airfoil. Ramamurti and Sandberg [6] numerically simulated unsteady viscous flow around oscillating NACA 0012 airfoil. They showed a good agreement between their numerical results and experimental data in terms of aerodynamics forces. Tay et al. [7] carried out a simulation to assess the lift, thrust and propulsive efficiency of different types of non-symmetrical airfoils under different flapping configurations. Their results showed that the shape of the airfoil has a profound effect on the lift, thrust, and efficiency. Boudis et al. [8] studied the flow structures and the forces acting on NACA0012 airfoil and they concluded that the trajectory motion has a significant effect on the performances and the vortex-shedding mode of the flapping airfoil.

Previous research in the area of the flapping airfoils showed the importance of understanding the flapping airfoil aerodynamics to improve airfoil design tools for unmanned aerial vehicles (UAVs) and micro aerial vehicles (MAVS). Therefore, this study is focusing on the aerodynamics of a number of the flapping birds' airfoils and one man made bird-like airfoil. The study is carried out for two types of birds' airfoils (Eagle, and Owl) and one man made bird-like airfoil S1223. The development of the pressure contours on upper and lower surfaces of the airfoil during the downstroke and upstroke modes is studied in details. Also, the periodic change of the lift and drag during the flapping motion were analyzed.

II. Governing Equations and Numerical Set Up

A. Governing Equations

In this research, the ANSYS Fluent is used to simulate the air flow around the flapping airfoils. The Reynolds Averaged Navier Stokes (RANS) equations are used. The flowfield is assumed to be fully turbulent, and k-epsilon turbulence model is implemented. The RANS equations and the formulation of the k-epsilon turbulence model can be found in reference [9, 10]. The dynamic mesh model in ANSYS Fluent is used to flap the airfoil where the pitching domain is changing with time due to motion on the domain boundaries. The update of

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the volume mesh is handled automatically by ANSYS Fluent at each time step based on the new positions of the boundaries [11].

B. Numerical Set Up

The airfoils of the owl and the eagle were selected to be simulated at the same Reynolds number and angular speed. The airspeed and flapping speed were set to 10 m/s and ±0.5 rad/s, respectively throughout the calculations. The Reynolds number was estimated as $7x10^5$. To resolve the viscous region, the yplus value was kept below 100. The simulation procedure followed was divided into three stages: Pre-processing, solver, and post process. The following section is a brief explanation of the three stages:

1- Pre-Processing

The pre-processing is started by importing the owl, eagle and S1223 airfoil coordinates data [1, 2, 3]. Figure 1 shows the sketches of the three airfoils.

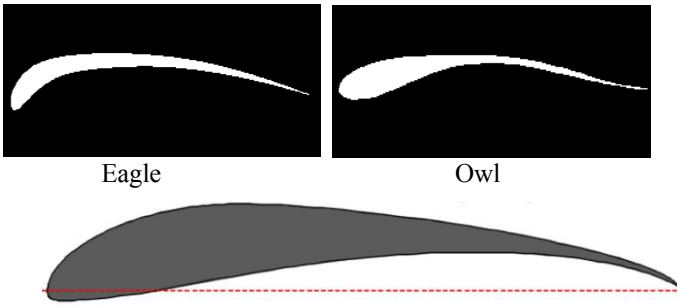

Fig. 1. Airfoils geometry

The C-H mesh type was generated using the ANSYS meshing module. The mesh domain was set distance between the airfoil center to 6 chords (6c) for the inlet and 12c for the outlet. The far boundary was located 6c away from the chord line. The clustered rotating mesh zone was drawn as a circle of 1.5c diameter. The element size of 0.02m was used but for the clustered rotating mesh zone the element size was reduced to 0.01m. The total elements of the mesh is found to be 71,208. Figure 2 shows the mesh views. The quality of the mesh was checked through orthogonal and skewness ratios. Good mesh quality was achieved. Also, the yplus distribution around airfoil was found to be in the acceptable range.

2- Solver

The transient viscous flow simulations were carried out. The k-epsilon-Realizable-Standard wall function turbulence model was used. The moving wall boundary with rotational motion was selected, and rotational axis was located at the center of the airfoil. No slip condition was applied to the airfoil surface. The rotational speed of 0.5 rad/s was used for downstroke and -0.5 rad/s for the up stroke. The 2nd order accuracy was used for spatial discretization of the pressure, momentum, turbulence kinetic energy and dissipation rate. The convergence criteria were set 10^{-4} and lift coefficient (Cl) and drag coefficient (Cd) time histories were plotted and stored.

The time step size was set to 0.005s with 400 total number of the time steps while the maximum iteration per time step of 20 was used. The converged history sample is shown in Figure 3.

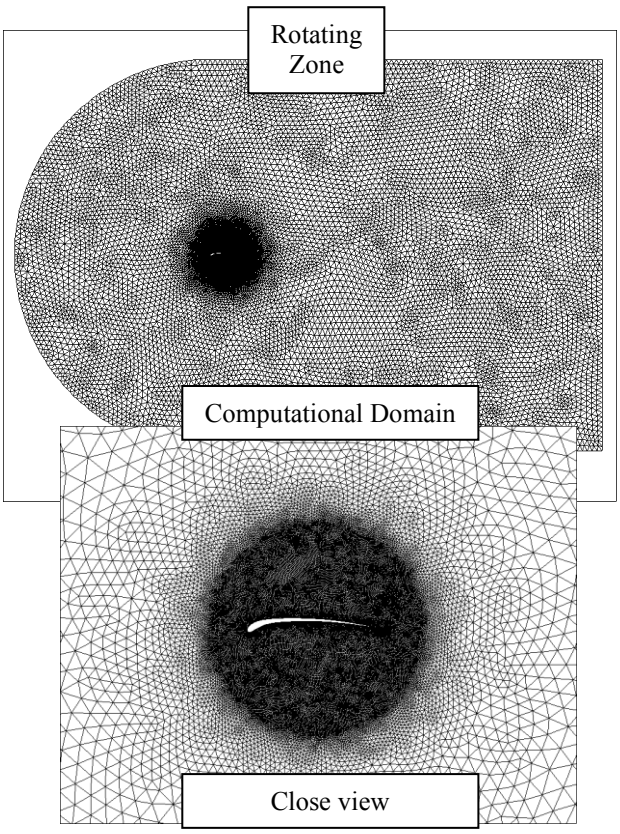

Fig. 2. Views of flapping airfoil mesh

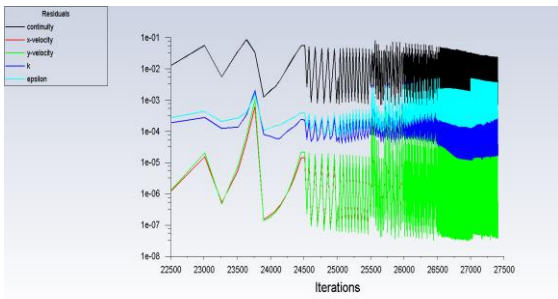

Fig. 3. Converging histories

III. RESULTS AND DISCUSSION

In this section, the numerical results are discussed in terms of the pressure contours, streamlines, and the time fluctuations of the lift and drag coefficients for the tested flapped airfoils.

A. Validation

The trend of the time fluctuations of the lift and drag coefficient curves was compared with the published data of other researchers [12-15]. A sample of the compared curves is plotted side by side with obtained curves from the present study as shown in Figure 4. The comparison revealed a good agreement between the present study and the literature in terms of the drag and lift coefficient time fluctuations trend.

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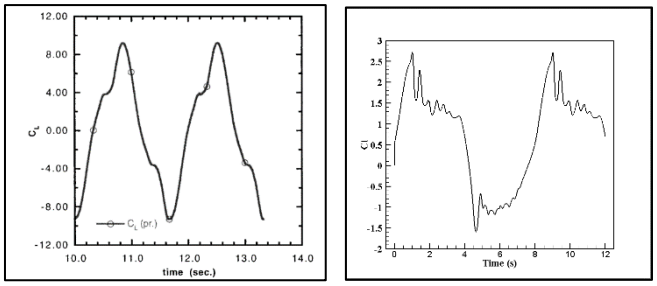
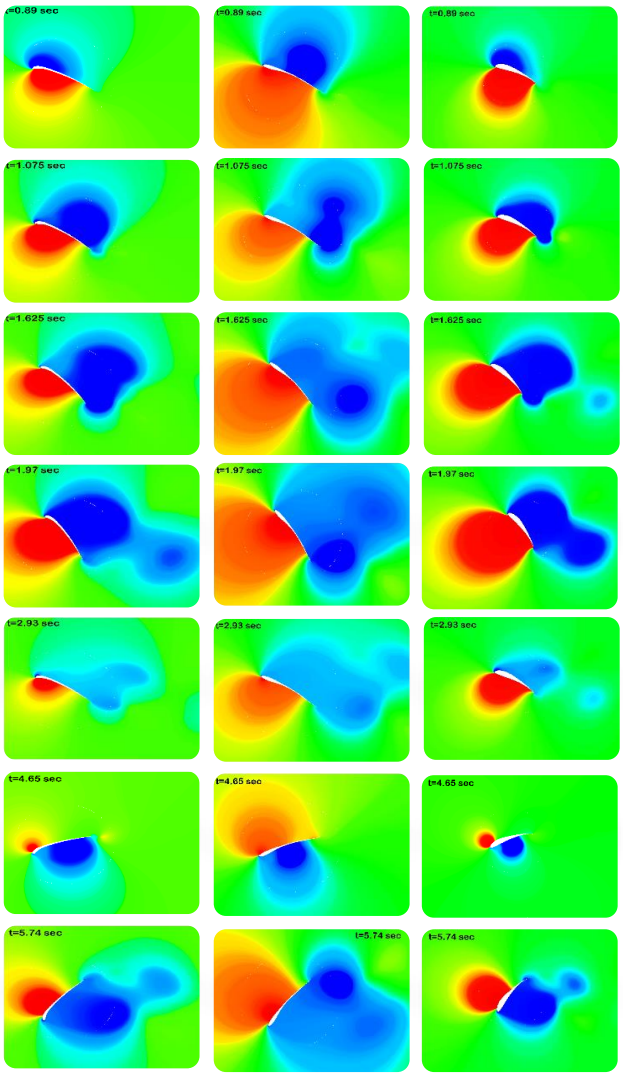


NACA0012 [15] Eagle

Fig. 4. Time fluctuations of the lift coefficient

B. Pressure Contours

Figure 5 compares the time history of the pressure contours for the three studied airfoils. The pressure develops in the same manner for the downstroke and upstroke. The downstroke period ends at time of 2 seconds where the airfoil enters the upstroke mode for a period of 4 seconds reaching the maximum rotation angle. The upstroke mode period last for 4 seconds after flapping back in downstroke mode for another 4 seconds. It is clear from Figure 5 that the size and strength of the three airfoils wake are not similar. In general, the appearance of the vortices shedding at the 2nd stage of the downstroke (t=1 to 2s) explains the fluctuation of the lift coefficient at that period as shown in Figure 7. The airfoil starts to flap upward to original position, two seconds are required for the airfoil to reach the starting position. Slowly, the vortices start to disappear, and the flow returns to its original pattern. As the airfoil moved to the up-stroke position, the vortex starts to appear on the lower surface of the airfoil and high-pressure region occupies the upper surface of the airfoil. Time 4.65s and 5.74s are presenting the upstroke flapping mode, the low pressure on the lower surface of the airfoil is clearly shown in Figure 5 for both times where the vortex is shredded to the wake region which is originally generated from the lower surface. This explains the negative lift coefficient and fluctuations shown in Figure 7. As the airfoil flaps back to the starting position, the vortex starts to weaken where the lift starts to increase. This scenario is repeated again for the 2nd downstroke from time 8 to 10s. Similarly, for the 2nd upstroke, the flow is behaving in the same manner as the first upstroke. Close view of each airfoil pressure contours development shows the appearance of the large and weak vortex on the upper surface of the owl airfoil at time of 0.89s where the low-pressure region is extended behind the airfoil trailing edge while at that period of time the eagle and S1223 airfoils found to be under the influence of strong vortex. As the flapping progresses to 1.075s, the eagle and S1223 airfoil pressure contours still develop in a similar trend where the low-pressure region is extended to the trailing edge but for the owl airfoil the vortex shedding behavior is observed earlier. One vortex moved away from the owl airfoil surface and the 2nd one still close to the surface but progressing to the wake region, but the strength of vortex seems to be weaker compared to the vortex on eagle and S1223 airfoils. For the rest of the downstroke period, the vortex shedding behavior appeared on all the airfoils. In summary, the S1223 airfoil showed a strong vortex in comparison to eagle and owl airfoils, however, the owl airfoil showed largest wake size. For the upstroke flapping mode still the owl airfoil generates a larger shedding region which is then followed by the eagle and S1223 airfoils.



Eagle Owl S1223

Fig. 5. Time history of the pressure contours

C. Streamlines

Figure 6 shows the development of the streamlines during the down and upstroke flapping modes. The vortex shedding developed in the early flapping period for the owl airfoil and divided into two less coherent vortices instead of one strong vortex and entering to the vortex shedding behavior. This observation supports the observation obtained from the pressure contour plots in Figure 5. At the end of the downstroke period of t=1.97s, the upper surface of the eagle airfoil is under the influence of the very strong vortex while for the owl airfoil the pair of weak vortices moved to the wake region in shedding behavior mode; and as for the S1223 airfoil one of the vortices is located close to the airfoil trailing edge. It is evident that the vortices size and strength generated by owl airfoil is smaller than the one generated by eagle airfoil. This will explain the silent flight feature of the owl airfoil. This scenario is repeated for the 2nd downstroke

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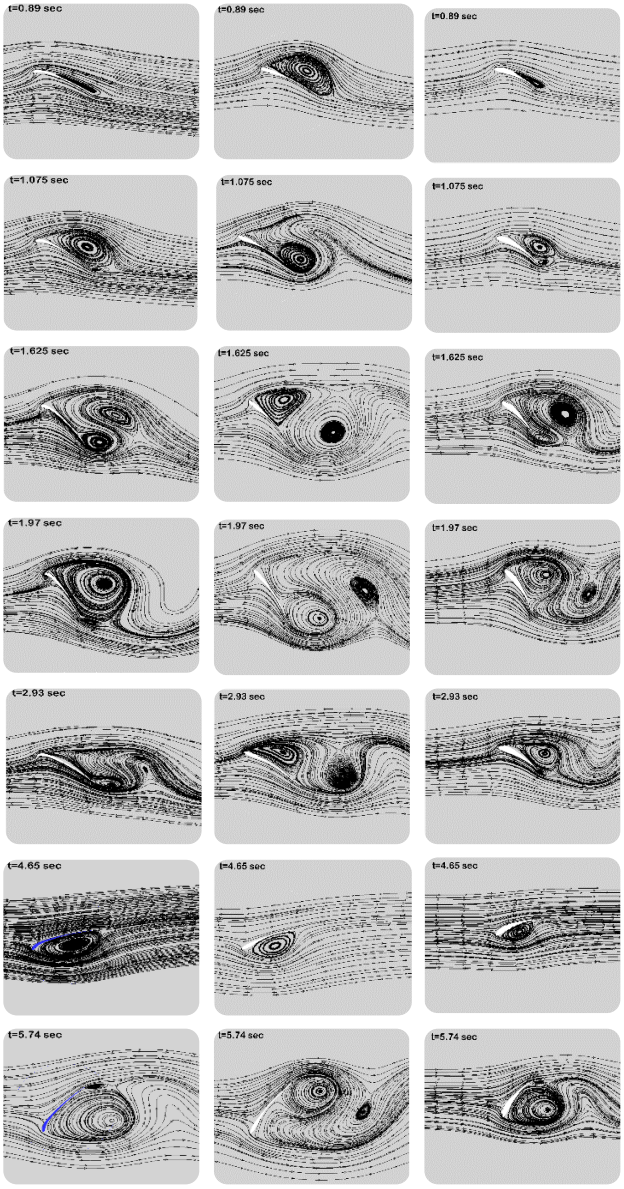
and upstroke where the flow is behaving in the same manner as 1st upstroke.



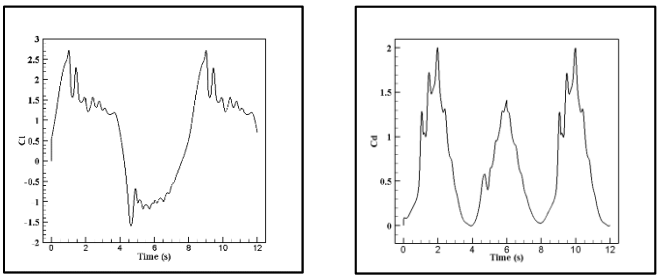
Eagle Owl S1223

Fig. 6. Time history of the streamlines

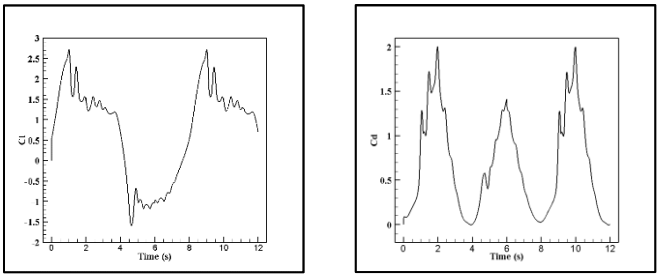
D. *Time History of the Lift and Drag Coefficients*

Figure 7 shows the time history of the lift and drag coefficient for the three studied airfoils. When the airfoil rotates downward (downstroke) reaching its maximum value at halfway after 1 second then start slightly to decrease and oscillating in the lift coefficient range of approximately 1.2 to 1.5. Those oscillations may be due to the unsteady vortex shedding behaviour at the wake region. This oscillating behaviour is maintained as the airfoil rotated upwards reaching to the starting position.

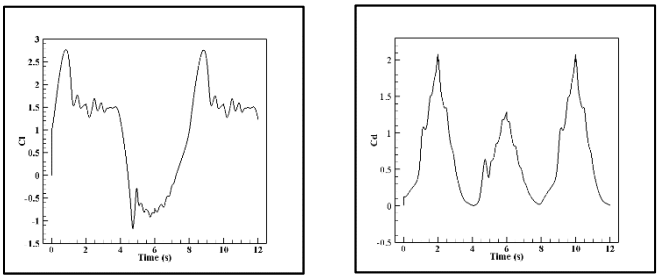
As the airfoil started the upstroke mode at a time of 4 seconds, suddenly the lift coefficient dropped to reach a negative value. This dropping of the lift is due to the existence of the low-pressure region on the lower surface of the airfoil and the high-pressure region on in the upper surface as shown the pressure contour plots in Figure 5. The minimum lift coefficient of -1.5, -2.2 and -1.2 were observed for the eagle, owl and S1223, respectively. Similarly, for the downstroke flapping mode, but this time the lift starts to increase slightly instead of decreasing with oscillations behaviour also start to appear. As the airfoil rotates downward to the original position, the lift coefficient increased rapidly but still had a negative value. As the airfoil enters the downstroke flapping mode for the 2nd time, the lift coefficient increases rapidly to reach its maximum again. Identical lift coefficient variation is observed for the 2nd flapping cycle.



Eagle



Owl



S1233

Fig. 7. The time of the lift and drag coefficients.

IV. CONCLUSION

In this study, the numerical simulations of the selected birds' airfoils (Eagle, Owl, and S1223) were carried out using Reynolds Averaged Navier Stokes Equations (RANS) equations where the effect of the turbulence was modelled using k-Epsilon Realized turbulence model. The results showed the growing in the wake region as the airfoil flapped. It is evident from the pressure contour plots that the strength and size of the vortices are dependent on the type of the airfoil and flapping time. The appearance of the large wake on the upper surface of the airfoil for downstroke flapping and on the lower surface for upstrokes flapping is observed. The strength

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and the size of the wake regions increase as the flapping angle increases. The larger wake region was observed for the eagle while a smaller one was observed for the owl airfoil. This will explain the silent flight feature of the owl wing. Also, this well-known silent feature of the owl flight may be due to the small variation of the pressure in the wake region for the owl airfoil in comparison to the other birds' airfoils. No thrust was observed for the studied airfoils since all the generated wake is classified as drag-indicative wake due to the selected speed and flapping rate. Good agreement with the published data was obtained in terms of the trend of the time fluctuation of the lift and drag coefficients. The current study can benefit the airfoil designer to understand the aerodynamic behavior of the flapping airfoil, explain the silent flight feature of the owl airfoil and adapt some of the bird's airfoil geometry features in designing flapping airfoils for unmanned or micro aerial vehicle.

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